

1. Functional Requirements

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Requirement

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R The system shall provide a concrete registration interface that an
- OpenClaw agent can invoke when delegating or starting background work.

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R The system shall persist tracked task state in a durable local store
- independent of conversational memory.

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R The system shall use a scheduler independent of agent context to evaluate
- active tasks at a configured interval.

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R The system shall represent task states using explicit definitions: running,
- completed, failed, stale, unknown.

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R The system shall emit proactive updates for completion, failure, and
- overdue-running states.

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R The system shall suppress duplicate notifications using explicit last-
- notification metadata.

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F The system shall record evidence of observed task state, emitted updates,
R and scheduler activity.

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R The system shall support configurable check-in intervals, with a target
- default of 2–3 minutes when feasible.

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R The system shall survive agent/session restart without losing tracked active
- tasks.

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R The system shall support Phase 1 task types only where observation is
- concretely defined; unsupported task types shall be marked unsupported
1 rather than silently treated as tracked.

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2. Non-Functional Requirements

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R The system shall not depend on a human manually requesting status.

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R The system shall distinguish observed, inferred, and unverified states in
its outputs.

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The system shall emit honest degraded-status updates instead of false

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- positives.

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The system shall be operable on the current Ubuntu/OpenClaw VM without major external infrastructure.

The system shall minimize routine token/compute overhead by keeping Phase 1 check logic deterministic and non-LLM.

3. Interface Requirement

Phase 1 registration interface: a local command or script with explicit parameters (task id/title/owner/interval/observable handle/target). This interface must be specified before implementation begins.